

Convex Optimization Project

November 20, 2023

General Description

You are asked to write a branch-and-bound algorithm for the binary knapsack problem. The problem can be easily described as follows. Given a set of n items, each with profit p_j and weight w_j , and a single container of capacity W , select a subset of the items of maximum total profit that fits into the container.

A MIP formulation and a greedy algorithm to solve its LP relaxation (you do **not** need an LP solver!) are described in Fischetti, Ch 8.0.3.

Input/Output

The code should read a problem from a text file with the following format:

```
n W
p_1 w_1
p_2 w_2
...
p_n w_n
```

and write the optimal solution to again to a text file with the following format:

```
z*
x*_1
x*_2
...
x*_n
```

where z^* is the optimal objective value and x^* is the optimal solution computed by your algorithm.

Computational evaluation

Random instances to test your method should be generated with the following procedure:

- First generate a spanning set of v pairs (w_k, p_k) . Each w_k is chosen uniformly at random in the range $[1, 1000]$ and the corresponding profit p_k is chosen uniformly at random in the range $[w_k + 95, w_k + 105]$.
- Those v pairs are normalized by dividing all numbers by $m + 1$.
- n items are generated by repeating the following process n times: pick a pair at random and multiply it by a random multiplier in $[1, m]$ (and round up the numbers).
- The capacity W of the knapsack is always obtained as $\lceil \sum_{j=1}^n w_j / 3 \rceil$.

For the experiment, use the following values:

- $v = 5$
- $m = 20$
- $n = \{50, 60, 70, 80, 90, 100\}$

For each combination of parameters, generate 5 different instances, and report the computational results of your method on those (#instances solved, runtime, number of nodes, final optimality gap at time limit reached). For each instance, use a time limit of 5 minutes.

Evaluation

Submit your code (allowed languages: C/C++/Java/Python) and a brief document describing what you did and reporting computational results by December 22nd, 2022. The possible score are:

score	description
+2	outstanding
+1	good
+0	mediocre
-1	unsatisfactory/buggy/cheating

Note that the project is optional, but if submitted must be individual: no group projects!